SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence   COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4   Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I _____BAIRAGANI DINESH 192372189______________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

I. If a patient has Fever AND Cough → Possible Flu

Ii. If a patient has Fever AND Rash → Possible Measles

Iii. If a patient has Cough AND Breathlessness → Possible Pneumonia

Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

I. ∀x: Vehicle(x) ∧ EmergencyType(x) → ClearPath(x)   [Rule 1]

II. ∀x: ClearPath(x) → GreenSignal(x) ∧ ¬RedSignal(x)   [Rule 2]

III. ∀x ∀y: Vehicle(x) ∧ Behind(x,y) ∧ GreenSignal(y) → Proceed(x)   [Rule 3]

IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

1. P1: SoilDry → IrrigationNeeded
2. P2: IrrigationNeeded ∧ CropWheat → ApplyDripMethod
3. P3: ¬ApplyDripMethod ∧ CropWheat → CropAtRisk
4. Facts: SoilDry = True, CropWheat = True

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication → move ¬ inward → distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause □. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

1. Stench [1,2] → Wumpus is adjacent to cell [1,2]
2. Breeze [1,1] → Pit is adjacent to cell [1,1]
3. Glitter [x,y] → Gold is at [x,y]
4. ¬Wumpus [x,y] ∧ ¬Pit[x,y] → Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] → [1,2] → [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided

Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

1ANS: Smart Medical Diagnosis System:

(a) Propositional Logic Representation

Definition of Propositional Symbols

Let the propositional variables be defined as follows:

- F: Patient A has Fever
- C: Patient A has Cough
- R: Patient A has Rash
- B: Patient A has Breathlessness
- FL: Patient A has Possible Flu
- M: Patient A has Possible Measles
- P: Patient A has Possible Pneumonia

Knowledge Base (KB) Representation

Rules:

1. **Rule 1 (R_1):** $(F \wedge C) \to FL$
2. **Rule 2 (R_2):** $(F \wedge R) \to M$
3. **Rule 3 (R_3):** $(C \wedge B) \to P$

Facts:

4. **Fact 1 (F_1):** F (True)
5. **Fact 2 (F_2):** C (True)
6. **Fact 3 (F_3):** B (True)

(b) Forward Chaining

Initial Known Facts: $\{F, C, B\}$

Iteration 1:

- **Check Rule 1 (R_1):** $(F \wedge C) \rightarrow FL$
    - Antecedents F and C are both in the known facts set ($F =$ True, $C =$ True).
    - **Rule Fired:** R_1
    - **Fact Derived:** FL (Possible Flu)
    - **Updated Known Facts:** $\{F, C, B, FL\}$
- **Check Rule 3 (R_3):** $(C \wedge B) \rightarrow P$
    - Antecedents C and B are both in the known facts set ($C =$ True, $B =$ True).
    - **Rule Fired:** R_3
    - **Fact Derived:** P (Possible Pneumonia)
    - **Updated Known Facts:** $\{F, C, B, FL, P\}$
- **Check Rule 2 (R_2):** $(F \wedge R) \rightarrow M$
    - F is known, but R (Rash) is **not** in the known facts set.
    - Rule R_2 cannot fire.

Final Derived Diagnoses:

- **Possible Flu (FL)**
- **Possible Pneumonia (P)**

(c) Backward Chaining & Goal Reduction Tree

Goal: Confirm P (Patient A has Pneumonia)

Step-by-Step Backward Chaining Process:

1. **Goal:** P
    - Find a rule with P in its conclusion: **Rule 3 (R_3).**
    - Rule 3 states: $(C \wedge B) \rightarrow P$.
    - Sub-goals generated: Verify C AND Verify B.
2. **Sub-goal 1:** Verify C
    - Check Knowledge Base: C exists directly as Fact 2 (F_2).
    - **Sub-goal C is satisfied.**
3. **Sub-goal 2:** Verify B

- Check Knowledge Base: B exists directly as Fact 3 (F_3).
- **Sub-goal B is satisfied.**

4. **Conclusion:** Both antecedents C and B are proven true via facts, confirming that P is **True**.

Goal Reduction Tree

Plaintext

```
      [ Goal: P ]  (Possible Pneumonia)
          |
       (Rule R3)
          |
    +-------+-------+
    |       |
 [ Sub-goal: C ]  [ Sub-goal: B ]
    |       |
  (Fact 2)    (Fact 3)
    |       |
  [ True ]    [ True ]
```

2 ans:-

(a) Unification

Unify Rule 1 (Vehicle(x) ∧ EmergencyType(x) → ClearPath(x)) with the facts:

1. **Match** Vehicle(x) **with** Vehicle(Ambulance)**:**

$$\theta_1 = \{x \mapsto \text{Ambulance}\}$$

2. **Match** EmergencyType(Ambulance) **with** EmergencyType(Ambulance)**:**

$$\theta_2 = \{\ \ \}$$

- **Most General Unifier (MGU):** $\theta = \{x \mapsto \text{Ambulance}\}$
- **Derived Fact:** ClearPath(Ambulance)

(b) Resolution Proof for Proceed(CarA)

CNF Conversion

1. $\neg$Vehicle(x) $\lor$ $\neg$EmergencyType(x) $\lor$ ClearPath(x)
2. $\neg$ClearPath(x) $\lor$ GreenSignal(x)
3. $\neg$Vehicle(x) $\lor$ $\neg$Behind(x, y) $\lor$ $\neg$GreenSignal(y) $\lor$ Proceed(x)
4. Vehicle(Ambulance)
5. EmergencyType(Ambulance)
6. Vehicle(CarA)
7. Behind(CarA,Ambulance)
8. $\neg$Proceed(CarA) *(Negated Goal)*

Resolution Steps

1. **Resolve 8 & 3** ($x \mapsto$ CarA):

$$\neg\text{Vehicle(CarA)} \lor \neg\text{Behind}(\text{CarA}, y) \lor \neg\text{GreenSignal}(y)$$

2. **Resolve with 6:**

$$\neg\text{Behind}(\text{CarA}, y) \lor \neg\text{GreenSignal}(y)$$

3. **Resolve with 7** ($y \mapsto$ Ambulance):

$$\neg\text{GreenSignal(Ambulance)}$$

4. **Resolve with 2** ($x \mapsto$ Ambulance):

$$\neg\text{ClearPath(Ambulance)}$$

5. **Resolve with 1** ($x \mapsto$ Ambulance):

$$\neg\text{Vehicle(Ambulance)} \lor \neg\text{EmergencyType(Ambulance)}$$

6. **Resolve with 4:**

$$\neg\text{EmergencyType(Ambulance)}$$

7. **Resolve with 5:**

□ *(Empty Clause — Goal Proved)*

(c) Analysis for Two Emergency Vehicles

Sufficient? No.

Missing Elements:

- **Conflict Resolution:** No priority rules exist if two emergency vehicles approach an intersection simultaneously from different directions.
- **Mutual Exclusion:** No rule prevents giving a green signal to perpendicular lanes at the same time.

Needed Rules:

- **Priority Rule:**

$$\forall x \forall y: \text{Emergency}(x) \land \text{Emergency}(y) \land \text{HigherPriority}(x, y) \rightarrow \text{RightOfWay}(x)$$

- **Safety Rule:**

$$\forall x \forall y: \text{Intersecting}(x, y) \land \text{GreenSignal}(x) \rightarrow \neg\text{GreenSignal}(y)$$

3ANS:-

Agricultural Expert Reasoning System:

(a) CNF Conversion

Symbols Used:

- SD: SoilDry
- IN: IrrigationNeeded
- CW: CropWheat
- ADM: ApplyDripMethod
- CAR: CropAtRisk

Step-by-Step Conversion:

1. **Rule P_1:** $SD \rightarrow IN$
    - **Eliminate →:** $\neg SD \lor IN$
    - **CNF 1:** $\neg SD \lor IN$
2. **Rule P_2:** $(IN \land CW) \rightarrow ADM$
    - **Eliminate →:** $\neg(IN \land CW) \lor ADM$
    - **Move ¬ inward (De Morgan's):** $\neg IN \lor \neg CW \lor ADM$
    - **CNF 2:** $\neg IN \lor \neg CW \lor ADM$
3. **Rule P_3:** $(\neg ADM \land CW) \rightarrow CAR$

- **Eliminate →:** $\neg(\neg ADM \wedge CW) \vee CAR$
- **Move ¬ inward (De Morgan's):** $ADM \vee \neg CW \vee CAR$
- **CNF 3:** $ADM \vee \neg CW \vee CAR$

4. **Facts:**
    - **CNF 4:** SD
    - **CNF 5:** CW

(b) Resolution Refutation for ApplyDripMethod (ADM)

Knowledge Base Clauses:

1. C_1: $\neg SD \vee IN$
2. C_2: $\neg IN \vee \neg CW \vee ADM$
3. C_3: $ADM \vee \neg CW \vee CAR$
4. C_4: SD
5. C_5: CW
6. C_6 (Negated Goal): $\neg ADM$

Resolution Steps:

1. **Resolve C_6 ($\neg ADM$) and C_2 ($\neg IN \vee \neg CW \vee ADM$):**

→ C_7: $\neg IN \vee \neg CW$

2. **Resolve C_7 ($\neg IN \vee \neg CW$) and C_5 (CW):**

→ C_8: $\neg IN$

3. **Resolve C_8 ($\neg IN$) and C_1 ($\neg SD \vee IN$):**

→ C_9: $\neg SD$

4. **Resolve C_9 ($\neg SD$) and C_4 (SD):**

→ ◻ *(Empty Clause)*

Resolution Proof Tree

Plaintext

```
[ C6: ¬ADM ]     [ C2: ¬IN ∨ ¬CW ∨ ADM ]
```

```
 \              /
  \            /
 [ C7: ¬IN ∨ ¬CW ]     [ C5: CW ]
       \              /
        \            /
        [ C8: ¬IN ]      [ C1: ¬SD ∨ IN ]
             \              /
              \            /
              [ C9: ¬SD ]      [ C4: SD ]
                   \          /
                    \        /
                    [ Empty □ ]
```

(c) Effect of Removing SoilDry

If SD is removed from the Knowledge Base:

1. **Can we infer IN (**IrrigationNeeded**)?**
   - **No.** P_1 ($\neg SD \vee IN$) requires $SD =$ True via Modus Ponens to derive IN. Without SD, no rule fires to prove IN.
2. **Can we infer ADM (**ApplyDripMethod**)?**
   - **No.** Deriving ADM requires both IN and CW to be true. Since IN is unknown, ADM cannot be derived.
3. **Does CropAtRisk (CAR) follow?**
   - **No.** Under Classical Logic, neither ADM nor $\neg ADM$ is known, so P_3 cannot fire to prove CAR.

Summary:

- Without the fact SD, the agent enters an **uncertain state** where neither ADM nor CAR logically follows from the Knowledge Base.

4ANS:

Wumpus World Knowledge Agent:

(a) Agent's Belief State & Modus Ponens Derivations

Initial Percept Facts (Given):

1. $S_{1,2}$: Stench at $[1,2]$
2. $B_{1,1}$: Breeze at $[1,1]$
3. $G_{2,2}$: Glitter at $[2,2]$

Modus Ponens Derivations:

1. **Rule 1:** $S_{1,2} \rightarrow (W_{1,1} \vee W_{2,2} \vee W_{1,3})$
   - Applying **Modus Ponens** with $S_{1,2}$:
   - **Derived:** $W_{1,1} \vee W_{2,2} \vee W_{1,3}$ *(Wumpus is at* $[1,1]$*,* $[2,2]$*, or* $[1,3]$*)*
2. **Rule 2:** $B_{1,1} \rightarrow (P_{1,2} \vee P_{2,1})$
   - Applying **Modus Ponens** with $B_{1,1}$:
   - **Derived:** $P_{1,2} \vee P_{2,1}$ *(Pit is at* $[1,2]$ *or* $[2,1]$*)*
3. **Rule 3:** $G_{2,2} \rightarrow Gold_{2,2}$
   - Applying **Modus Ponens** with $G_{2,2}$:
   - **Derived:** $Gold_{2,2}$ *(Gold is at* $[2,2]$*)*

List of Derived Conclusions:

- **Gold:** Located at $[2,2]$.
- **Wumpus:** Located in one of $\{[1,1], [2,2], [1,3]\}$.
- **Pit:** Located in one of $\{[1,2], [2,1]\}$.

(b) Possible Locations of Wumpus and Pit (Forward Chaining)

Adjacent Grid Map (Standard 2D Grid):

- **Adjacent to** $[1,2]$**:** $\{[1,1], [1,3], [2,2]\}$

- **Adjacent to** $[1,1]$**:** $\{[1,2],[2,1]\}$

Step-by-Step Reasoning:

1. **Step 1 (Wumpus Location):**
   - Percept $S_{1,2}$ fires Rule 1.
   - Forward chaining concludes: Wumpus ∈ $\{[1,1],[2,2],[1,3]\}$.
   - Since the agent safely navigated $[1,1]$ to feel the breeze without dying, $[1,1]$ contains no Wumpus ($\neg W_{1,1}$).
   - **Possible Wumpus Locations:** $\{[2,2],[1,3]\}$
2. **Step 2 (Pit Location):**
   - Percept $B_{1,1}$ fires Rule 2.
   - Forward chaining concludes: Pit ∈ $\{[1,2],[2,1]\}$.
   - **Possible Pit Locations:** $\{[1,2],[2,1]\}$

(c) Safety Evaluation of Path $[\mathbf{1},\mathbf{1}] \rightarrow [\mathbf{1},\mathbf{2}] \rightarrow [\mathbf{2},\mathbf{2}]$

Goal: Determine if Safe$[1,2]$ and Safe$[2,2]$ hold.

Rule 4: $\neg W_{x,y} \land \neg P_{x,y} \rightarrow$ Safe$[x,y]$

Evaluation of Cell $[\mathbf{1},\mathbf{2}]$:

- From Step 2 in part (b), a Pit exists in either $[1,2]$ or $[2,1]$ ($P_{1,2} \lor P_{2,1}$).
- $P_{1,2}$ cannot be proven false ($\neg P_{1,2}$ is not established).
- **Conclusion for** $[1,2]$**:** Cell $[1,2]$ carries a risk of containing a Pit ($P_{1,2}$). Therefore, Safe$[1,2]$ **cannot be guaranteed**.

Evaluation of Cell $[\mathbf{2},\mathbf{2}]$:

- From Step 1 in part (b), the Wumpus could be at $[2,2]$ ($W_{2,2} \lor W_{1,3}$).
- $W_{2,2}$ cannot be proven false ($\neg W_{2,2}$ is not established).
- **Conclusion for** $[2,2]$**:** Cell $[2,2]$ carries a risk of containing the Wumpus.

Final Judgment:

The path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ **is NOT safe.**

- Cell $[1,2]$ has a potential **Pit**.
- Cell $[2,2]$ has a potential **Wumpus**.
- The agent cannot satisfy Rule 4 ($\neg W \wedge \neg P$) for either cell on this path